



6987 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	17	MIRI MRS centered on EtaCar-E	MIRI Medium Resolution Spectroscopy	(9) ETA-CAR-MIRI-E
	18	MIRI MRS centered on EtaCar-W	MIRI Medium Resolution Spectroscopy	(10) ETA-CAR-MIRI-W
	7	MIRI MRS Background E	MIRI Medium Resolution Spectroscopy	(4) ETACAR-BACK-E

Folder	Observation	Label	Observing Template	Science Target
	22	MIRI MRS Background ODF	MIRI Medium Resolution Spectroscopy	(14) ETACAR-BACK-ODF
	23	MIRI MRS Background W	MIRI Medium Resolution Spectroscopy	(13) ETACAR-BACK-W
	11	MIRI MRS ODF E5	MIRI Medium Resolution Spectroscopy	(5) ODF-E5-MIRI
	2	NIRSPEC IFU ODF E5	NIRSpec IFU Spectroscopy	(15) ODF-E5-NIRSpec
	3	NIRSPEC IFU ODF COND	NIRSpec IFU Spectroscopy	(6) ODF-S-COND
	1	NIRSPEC IFU BACK	NIRSpec IFU Spectroscopy	(16) ETACAR-BACK-E-NIRspec
	19	NIRSpecIFU SE mosaic	NIRSpec IFU Spectroscopy	(11) SE-LOBE-NIRSpec-Mosaic
	20	NIRSpecIFU NW mosaic	NIRSpec IFU Spectroscopy	(12) NW-LOBE-NIRSpec-mosaic
	21	FS Mosaic center/butterfly	NIRSpec Fixed Slit Spectroscopy	(1) -ETA-CAR
	31	FS Mosaic center/butterfly - Repeat of obs 21	NIRSpec Fixed Slit Spectroscopy	(1) -ETA-CAR
	41	FS Mosaic center/butterfly - Repeat of obs 31	NIRSpec Fixed Slit Spectroscopy	(1) -ETA-CAR

ABSTRACT

The Luminous Blue Variable Eta Carinae is a colliding-wind binary, the most massive stellar multiple within 3 kpc, and a strong candidate for a merger in a born-triple system. Its spectacular bipolar 10" x 20" Homunculus Nebula, formed in the "Great Eruption" in the 1840s, is postulated to have been ejected in a super-Eddington wind during the spiral-in phase of two merging stars. The nitrogen-rich nebula contains at least 0.4 Msun of dust, with a diverse chemistry reflecting the CNO-processed abundance pattern of the erupting star's surface. However, the mass and chemical distribution of the dust between the polar lobes and a warm wind-carved equatorial torus -- tracers of the system's mass-loss history -- are highly uncertain since the only spectrum revealing the unique dust bands is spatially unresolved. We propose to decisively fill this crucial gap in our knowledge with JWST spectral maps of the nebula and two positions in the nearby debris field containing N-rich material ejected centuries before the Great Eruption, possibly during an unstable triple orbit phase involving mass-stripping encounters. The dust bands and nebular lines provide essential diagnostics of the distribution of dust species, grain properties, and associated excitation conditions, and will constrain the dynamical models of the system with the mass budget over its eruptive history, and provide a detailed understanding of the photon- and shock-driven gas excitation. The science unique to JWST's capabilities will offer insights into key processes like common-envelope evolution, a critical yet poorly-

understood phase in the formation of gravitational wave sources.

OBSERVING DESCRIPTION

We propose to perform NIRSpec and MIRI MRS IFU spectroscopy on the dusty 10" x 20" Homunculus Nebula and nearby debris fields surrounding the Luminous Blue Variable and candidate merger-in-a-triple eta Carinae. The observations on the Homunculus are designed as mosaics providing maximum wavelength coverage over the full nebula to separate the various anticipated dust bands, nebular and recombination lines in the complex geometry of the disrupted equatorial torus, the equatorial skirt, and bipolar lobes to characteristic clumping scales of $\sim 0.2''$. Two positions target pre-Great Eruption material in the surrounding debris field for spectroscopy with a single pointing of the IFUs.

The NIRSpec G140H/F100LP, G235H/170LP, and G395H/290LP disperser-filter combinations are used to cover emission from hot dust, nebular and reflected recombination H and He, fine structure lines, and ro-vibrational H₂ which will trace photon- and shock-excited material in the Homunculus and X-ray bright ring of the outer ejecta. The plentiful spectral lines will prevent IFU wavelength gaps from interfering with our science goals. High spectral resolution will also moderate throughput from the bright central region of the Homunculus, avoiding saturation according to our sensitivity estimates with ETC 4.0. MSA leak-cals are included with the NIRSpec IFU science observations.

The MIRI MRS observations employ all 4 channels to similarly cover warm and cooler dust species across the nebula, H and He lines, and key fine-structure lines to trace where molecule and dust formation is inhibited. Group numbers are set to 6, minimizing persistence and saturation risks based on ETC 4.0 sensitivity calculations, ground-based imaging between 0.658 and 19.5 microns, and SED shape indicated by spatially-unresolved ISO spectroscopy of the Homunculus.

Proposal 6987 - Targets - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	-ETA-CAR	RA: 10 45 3.5378 (161.2647408d) Dec: -59 41 4.05 (-59.68446d) Equinox: J2000	Proper Motion RA: -11 mas/yr Proper Motion Dec: 4.1 mas/yr Epoch of Position: 2016.0	
	Comments: Updated to use Gaia DR3 position. Gaia DR3 does not provide any proper motions for this target, so we have adopted UCAC4 proper motion but with the Gaia DR3 position at 2016.0 as the reference. The UCAC4 motion looks real, but it isn't clear if it was measuring the central star or some other ejected material. The uncertainty as to whether or not to include the proper motion probably adds ~ 0.1" to the final positional uncertainty. Category=Star Description=[Eta Carinae stars] Extended=YES				
	(4)	ETACAR-BACK-E	RA: 10 43 58.6400 (160.9943333d) Dec: -59 47 20.22 (-59.78895d) Equinox: J2000	Epoch of Position: 2000.0	
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES				
	(5)	ODF-E5-MIRI	RA: 10 45 7.0200 (161.2792500d) Dec: -59 41 12.30 (-59.68675d) Equinox: J2000	Proper Motion RA: 50.0 mas/yr Proper Motion Dec: -19.0 mas/yr Epoch of Position: 2000.0	
	Comments: Category=ISM Description=[High-velocity clouds] Extended=YES				
	(6)	ODF-S-COND	RA: 10 45 2.4000 (161.2600000d) Dec: -59 41 7.30 (-59.68536d) Equinox: J2000	Proper Motion RA: -48.0 mas/yr Proper Motion Dec: -6.0 mas/yr Epoch of Position: 2000	
	Comments: Category=ISM Description=[High-velocity clouds] Extended=YES				
	(9)	ETA-CAR-MIRI-E	RA: 10 45 4.0240 (161.2667667d) Dec: -59 41 7.36 (-59.68538d) Equinox: J2000	Proper Motion RA: -0.0014528297321410096 sec of time/yr Proper Motion Dec: 0.0040999999999999995 arcsec/yr Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Eta Carinae stars] Extended=YES					
	(10)	ETA-CAR-MIRI-W	RA: 10 45 3.0680 (161.2627833d) Dec: -59 41 0.39 (-59.68344d) Equinox: J2000	Proper Motion RA: -0.0014528297321410096 sec of time/yr Proper Motion Dec: 0.0040999999999999995 arcsec/yr Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Eta Carinae stars] Extended=YES					

Proposal 6987 - Targets - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

(11)	SE-LOBE-NIRSpec-Mosaic	RA: 10 45 3.7098 (161.2654575d) Dec: -59 41 5.01 (-59.68473d) Equinox: J2000	
Comments: Category=ISM Description=[Emission nebulae] Extended=YES			
(12)	NW-LOBE-NIRSpec-mosaic	RA: 10 45 3.3741 (161.2640588d) Dec: -59 41 2.97 (-59.68416d) Equinox: J2000	
Comments: Category=ISM Description=[Emission nebulae] Extended=YES			
(13)	ETACAR-BACK-W	RA: 10 43 58.6400 (160.9943333d) Dec: -59 47 20.22 (-59.78895d) Equinox: J2000	Epoch of Position: 2000.0
Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES			
(14)	ETACAR-BACK-ODF	RA: 10 43 58.6400 (160.9943333d) Dec: -59 47 20.22 (-59.78895d) Equinox: J2000	Epoch of Position: 2000.0
Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES			
(15)	ODF-E5-NIRSpec	RA: 10 45 7.0200 (161.2792500d) Dec: -59 41 12.30 (-59.68675d) Equinox: J2000	Proper Motion RA: 50.0 mas/yr Proper Motion Dec: -19.0 mas/yr Epoch of Position: 2000.0
Comments: Category=ISM Description=[High-velocity clouds] Extended=YES			
(16)	ETACAR-BACK-E-NIRspec	RA: 10 43 58.6400 (160.9943333d) Dec: -59 47 20.22 (-59.78895d) Equinox: J2000	Epoch of Position: 2000.0
Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES			

Proposal 6987 - Observation 17 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 17: MIRI MRS centered on EtaCar-E					Tue May 26 15:00:26 GMT 2026	
	Diagnostic Status: Warning						
	Observing Template: MIRI Medium Resolution Spectroscopy						
	Background Observations:[MIRI MRS Background E (Obs 7)]						
Diagnostics	(MIRI MRS centered on EtaCar-E (Obs 17)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.						
	(MIRI MRS centered on EtaCar-E (Obs 17)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.						
	(MIRI MRS centered on EtaCar-E (Obs 17)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.						
	(Visit 17:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.						
Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections		Miscellaneous	
	(9)	ETA-CAR-MIRI-E	RA: 10 45 4.0240 (161.2667667d) Dec: -59 41 7.36 (-59.68538d) Equinox: J2000	Proper Motion RA: -0.0014528297321410096 sec of time/yr Proper Motion Dec: 0.004099999999999995 arcsec/yr Epoch of Position: 2015.5			
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.						
	Category=Star						
	Description=[Eta Carinae stars]						
Acquisition	Extended=YES						
	#	Target					
	1	NONE					
Template	AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray		Grating Wheel Direction	
		All MRS	YES	FULL		Allow Auto Reorder	
Mosaic	Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
	2	2	15.0	15.0	0.0	0.0	ROW_ORDER
Dithers	#	Dither Type		Optimized For		Direction	
	1	4-Point		EXTENDED SOURCE		NEGATIVE	

Proposal 6987 - Observation 17 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	3	1	1	Dither 1	4	4	33.3	
	1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
	1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
	2		IMAGER	F560W	FASTR1	3	1	1	Dither 1	4	4	33.3	
	2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
	2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
	3		IMAGER	F560W	FASTR1	3	1	1	Dither 1	4	4	33.3	
	3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
	3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
Special Requirements	Aperture PA Range 310 to 350 Degrees (V3 310.0 to 350.0)												
	Sequence Observations 7, 17, Non-interruptible												

Proposal 6987 - Observation 18 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 18: MIRI MRS centered on EtaCar-W					Tue May 26 15:00:26 GMT 2026
	Diagnostic Status: Warning					
	Observing Template: MIRI Medium Resolution Spectroscopy					
	Background Observations:[MIRI MRS Background W (Obs 23)]					
Diagnostics	(MIRI MRS centered on EtaCar-W (Obs 18)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.					
	(MIRI MRS centered on EtaCar-W (Obs 18)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.					
	(MIRI MRS centered on EtaCar-W (Obs 18)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.					
	(Visit 18:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.					
Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous	
	(10)	ETA-CAR-MIRI-W	RA: 10 45 3.0680 (161.2627833d) Dec: -59 41 0.39 (-59.68344d) Equinox: J2000	Proper Motion RA: -0.0014528297321410096 sec of time/yr Proper Motion Dec: 0.004099999999999995 arcsec/yr Epoch of Position: 2015.5		
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.					
	Category=Star					
	Description=[Eta Carinae stars]					
Acquisition	Extended=YES					
	#	Target				
	1	NONE				
Template	AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction	
		All MRS	YES	FULL	Allow Auto Reorder	
Mosaic	Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)
	2	2	15.0	15.0	0.0	0.0
Dithers						Tile Order
						ROW_ORDER
Dithers	#	Dither Type		Optimized For		Direction
	1	4-Point		EXTENDED SOURCE		NEGATIVE

Proposal 6987 - Observation 18 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	3	1	1	Dither 1	4	4	33.3	
	1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
	1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
	2		IMAGER	F560W	FASTR1	3	1	1	Dither 1	4	4	33.3	
	2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
	2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
	3		IMAGER	F560W	FASTR1	3	1	1	Dither 1	4	4	33.3	
	3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601	
	3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601	
Special Requirements	Aperture PA Range 37 to 80 Degrees (V3 37.0 to 80.0)												
	Sequence Observations 23, 18 (reordered), Non-interruptible												

Proposal 6987 - Observation 7 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 7: MIRI MRS Background E												Tue May 26 15:00:26 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [MIRI MRS centered on EtaCar-E (Obs 17)]													
Diagnostics	(MIRI MRS Background E (Obs 7)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.													
	(MIRI MRS Background E (Obs 7)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.													
	(MIRI MRS Background E (Obs 7)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.													
	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(4)	ETACAR-BACK-E	RA: 10 43 58.6400 (160.9943333d) Dec: -59 47 20.22 (-59.78895d) Equinox: J2000					Epoch of Position: 2000.0						
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
		All MRS				YES			FULL		Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/E xp	Exposures/Dit h	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	FASTR1	3	1	1	Dither 1	4	4	33.3		
	1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601		
	1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601		
	2		IMAGER	F560W	FASTR1	3	1	1	Dither 1	4	4	33.3		
	2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601		
	2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601		
	3		IMAGER	F560W	FASTR1	3	1	1	Dither 1	4	4	33.3		
	3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601		
	3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601		

Proposal 6987 - Observation 7 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Special Requirements	Sequence Observations 7, 17, Non-interruptible
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Proposal 6987 - Observation 22 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 22: MIRI MRS Background ODF												Tue May 26 15:00:26 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [MIRI MRS ODF E5 (Obs 11)]													
Diagnostics	(Visit 22:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(14)	ETACAR-BACK-ODF	RA: 10 43 58.6400 (160.9943333d) Dec: -59 47 20.22 (-59.78895d) Equinox: J2000				Epoch of Position: 2000.0							
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				YES			FULL			Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	FASTR1	5	1	1	Dither 1	4	4	55.501		
	1	LONG(C)	MRSLONG		FASTR1	10	1	1	Dither 1	4	4	111.002		
	1	LONG(C)	MRSSHORT		FASTR1	10	1	1	Dither 1	4	4	111.002		
	2		IMAGER	F560W	FASTR1	5	1	1	Dither 1	4	4	55.501		
	2	MEDIUM(B)	MRSLONG		FASTR1	10	1	1	Dither 1	4	4	111.002		
	2	MEDIUM(B)	MRSSHORT		FASTR1	10	1	1	Dither 1	4	4	111.002		
	3		IMAGER	F560W	FASTR1	5	1	1	Dither 1	4	4	55.501		
	3	SHORT(A)	MRSLONG		FASTR1	10	1	1	Dither 1	4	4	111.002		
	3	SHORT(A)	MRSSHORT		FASTR1	10	1	1	Dither 1	4	4	111.002		

Proposal 6987 - Observation 22 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Special Requirements	Group Observations 11, 22, Non-interruptible
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Proposal 6987 - Observation 23 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 23: MIRI MRS Background W												Tue May 26 15:00:26 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [MIRI MRS centered on EtaCar-W (Obs 18)]													
Diagnostics	(MIRI MRS Background W (Obs 23)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.													
	(MIRI MRS Background W (Obs 23)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.													
	(MIRI MRS Background W (Obs 23)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.													
	(Visit 23:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous				
	(13)	ETACAR-BACK-W		RA: 10 43 58.6400 (160.9943333d)			Epoch of Position: 2000.0							
				Dec: -59 47 20.22 (-59.78895d)										
				Equinox: J2000										
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
		All MRS			YES			FULL			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	FASTR1	3	1	1	Dither 1	4	4	33.3		
	1	SHORT(A)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601		
	1	SHORT(A)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601		
	2		IMAGER	F560W	FASTR1	3	1	1	Dither 1	4	4	33.3		
	2	MEDIUM(B)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601		
	2	MEDIUM(B)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601		
	3		IMAGER	F560W	FASTR1	3	1	1	Dither 1	4	4	33.3		
	3	LONG(C)	MRSLONG		FASTR1	6	1	1	Dither 1	4	4	66.601		
	3	LONG(C)	MRSSHORT		FASTR1	6	1	1	Dither 1	4	4	66.601		

Proposal 6987 - Observation 23 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Special Requirements	Sequence Observations 23, 18 (reordered), Non-interruptible
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Proposal 6987 - Observation 11 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 11: MIRI MRS ODF E5												Tue May 26 15:00:26 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MIRI MRS Background ODF (Obs 22)]												
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(5)	ODF-E5-MIRI	RA: 10 45 7.0200 (161.2792500d)					Proper Motion RA: 50.0 mas/yr					
			Dec: -59 41 12.30 (-59.68675d)					Proper Motion Dec: -19.0 mas/yr					
			Equinox: J2000					Epoch of Position: 2000.0					
		Comments: Category= <i>ISM</i> Description=[<i>High-velocity clouds</i>] Extended= <i>YES</i>											
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	1	LONG(C)	MRSLONG		FASTR1	10	1	1	Dither 1	4	4	111.002	
	1	LONG(C)	MRSSHORT		FASTR1	10	1	1	Dither 1	4	4	111.002	
	2		IMAGER	F560W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	2	MEDIUM(B)	MRSLONG		FASTR1	10	1	1	Dither 1	4	4	111.002	
	2	MEDIUM(B)	MRSSHORT		FASTR1	10	1	1	Dither 1	4	4	111.002	
	3		IMAGER	F560W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	3	SHORT(A)	MRSLONG		FASTR1	10	1	1	Dither 1	4	4	111.002	
	3	SHORT(A)	MRSSHORT		FASTR1	10	1	1	Dither 1	4	4	111.002	

Special Requirements	Group Observations 11, 22, Non-interruptible
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Proposal 6987 - Observation 2 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 2: NIRSPEC IFU ODF E5											Tue May 26 15:00:26 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Background Observations:[NIRSPEC IFU BACK (Obs 1)]											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(15)	ODF-E5-NIRSpec	RA: 10 45 7.0200 (161.2792500d) Dec: -59 41 12.30 (-59.68675d) Equinox: J2000				Proper Motion RA: 50.0 mas/yr Proper Motion Dec: -19.0 mas/yr Epoch of Position: 2000.0					
	Comments: Category=ISM Description=[High-velocity clouds] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	SPARSE-CYCLING		MEDIUM						1,2		
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	15	1	true	true	NONE	2	2	466.844	143984
	2	G140H/F100LP	NRSIRS2RAPID	15	1	false	true	NONE	2	2	466.844	143984
	3	G235H/F170LP	NRSIRS2RAPID	5	1	true	true	NONE	2	2	175.067	143984
	4	G235H/F170LP	NRSIRS2RAPID	5	1	false	true	NONE	2	2	175.067	143984
	5	G395H/F290LP	NRSIRS2RAPID	5	1	true	true	NONE	2	2	175.067	143984
	6	G395H/F290LP	NRSIRS2RAPID	5	1	false	true	NONE	2	2	175.067	143984

Proposal 6987 - Observation 2 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Special Requirements	Group Observations 1, 2, 3, Non-interruptible
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Proposal 6987 - Observation 3 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 3: NIRSPEC IFU ODF COND											Tue May 26 15:00:26 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(6)	ODF-S-COND	RA: 10 45 2.4000 (161.2600000d) Dec: -59 41 7.30 (-59.68536d) Equinox: J2000				Proper Motion RA: -48.0 mas/yr Proper Motion Dec: -6.0 mas/yr Epoch of Position: 2000					
	Comments: Category=ISM Description=[High-velocity clouds] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	SPARSE-CYCLING		MEDIUM						1,2		
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	15	1	true	true	NONE	2	2	466.844	143984
	2	G140H/F100LP	NRSIRS2RAPID	15	1	false	true	NONE	2	2	466.844	143984
	3	G235H/F170LP	NRSIRS2RAPID	5	1	true	true	NONE	2	2	175.067	143984
	4	G235H/F170LP	NRSIRS2RAPID	5	1	false	true	NONE	2	2	175.067	143984
	5	G395H/F290LP	NRSIRS2RAPID	5	1	true	true	NONE	2	2	175.067	143984
	6	G395H/F290LP	NRSIRS2RAPID	5	1	false	true	NONE	2	2	175.067	143984

Proposal 6987 - Observation 3 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Special Requirements	Group Observations 1, 2, 3, Non-interruptible
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Proposal 6987 - Observation 1 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 1: NIRSPEC IFU BACK											Tue May 26 15:00:26 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Background Observation For: [NIRSPEC IFU ODF E5 (Obs 2)]											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(16)	ETACAR-BACK-E-NIRspec	RA: 10 43 58.6400 (160.9943333d) Dec: -59 47 20.22 (-59.78895d) Equinox: J2000				Epoch of Position: 2000.0					
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	SPARSE-CYCLING		MEDIUM						1,2		
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	15	1	true	true	NONE	2	2	466.844	143984
	2	G140H/F100LP	NRSIRS2RAPID	15	1	false	true	NONE	2	2	466.844	143984
	3	G235H/F170LP	NRSIRS2RAPID	5	1	true	true	NONE	2	2	175.067	143984
	4	G235H/F170LP	NRSIRS2RAPID	5	1	false	true	NONE	2	2	175.067	143984
	5	G395H/F290LP	NRSIRS2RAPID	5	1	true	true	NONE	2	2	175.067	143984
	6	G395H/F290LP	NRSIRS2RAPID	5	1	false	true	NONE	2	2	175.067	143984

Proposal 6987 - Observation 1 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Special Requirements	Group Observations 1, 2, 3, Non-interruptible
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Proposal 6987 - Observation 19 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 19: NIRSpecIFU SE mosaic											Tue May 26 15:00:26 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 19:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(11)	SE-LOBE-NIRSpec-Mosaic	RA: 10 45 3.7098 (161.2654575d) Dec: -59 41 5.01 (-59.68473d) Equinox: J2000									
Template	Comments: Category=ISM Description=[Emission nebulae] Extended=YES											
	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	4	6	3.3		3.3		0.0		0.0		COLUMN_ORDER	
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	SPARSE-CYCLING			SMALL						1,2	
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	5	1	false	true	NONE	2	2	175.067	
	2	G140H/F100LP	NRSIRS2RAPID	5	1	true	true	NONE	2	2	175.067	
	3	G235H/F170LP	NRSIRS2RAPID	3	1	false	true	NONE	2	2	116.711	
	4	G235H/F170LP	NRSIRS2RAPID	3	1	true	true	NONE	2	2	116.711	
	5	G395H/F290LP	NRSIRS2RAPID	2	1	false	true	NONE	2	2	87.533	
	6	G395H/F290LP	NRSIRS2RAPID	2	1	true	true	NONE	2	2	87.533	

Proposal 6987 - Observation 19 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Special Requirements	Aperture PA Range 149.97164917 to 287.97164917 Degrees (V3 11.0 to 149.0) No Parallel Attachments Same Aperture PA 19, 20, 21 (V3 PAs differ)
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Proposal 6987 - Observation 20 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 20: NIRSpecIFU NW mosaic											Tue May 26 15:00:26 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 20:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(12)	NW-LOBE-NIRSpec-mosaic	RA: 10 45 3.3741 (161.2640588d) Dec: -59 41 2.97 (-59.68416d) Equinox: J2000									
	Comments: Category=ISM Description=[Emission nebulae] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	4	6	3.3		3.3		0.0		0.0		COLUMN_ORDER	
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	SPARSE-CYCLING			SMALL						1,2	
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	5	1	false	true	NONE	2	2	175.067	
	2	G140H/F100LP	NRSIRS2RAPID	5	1	true	true	NONE	2	2	175.067	
	3	G235H/F170LP	NRSIRS2RAPID	3	1	false	true	NONE	2	2	116.711	
	4	G235H/F170LP	NRSIRS2RAPID	3	1	true	true	NONE	2	2	116.711	
	5	G395H/F290LP	NRSIRS2RAPID	2	1	false	true	NONE	2	2	87.533	
	6	G395H/F290LP	NRSIRS2RAPID	2	1	true	true	NONE	2	2	87.533	

Proposal 6987 - Observation 20 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Special Requirements	No Parallel Attachments
	Same Aperture PA 19, 20, 21 (V3 PAs differ)

Proposal 6987 - Observation 21 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 21: FS Mosaic center/butterfly											Tue May 26 15:00:26 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 21:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	-ETA-CAR	RA: 10 45 3.5378 (161.2647408d) Dec: -59 41 4.05 (-59.68446d) Equinox: J2000				Proper Motion RA: -11 mas/yr Proper Motion Dec: 4.1 mas/yr Epoch of Position: 2016.0					
	Comments: Updated to use Gaia DR3 position. Gaia DR3 does not provide any proper motions for this target, so we have adopted UCAC4 proper mostion but with the Gaia DR3 position at 2016.0 as the reference. The UCAC4 motion looks real, but it isn't clear if it was measuring the central star or some other ejected material. The uncertainty as to whether or not to include the proper motion probably adds ~ 0.1" to the final positional uncertainty. Category=Star Description=[Eta Carinae stars] Extended=YES											
Template	TA Method		HFF Readout Mode				Slit			Subarray		
	NONE		false				S200A1			SUBS200A1		
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	3	18	10.0		0.0		0.0		0.0		ROW_ORDER	
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Ex p	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F070LP	S200A1	NRSRAPID	4	4	1	NONE	3	12	93.726	
	2	G235H/F170LP	S200A1	NRSRAPID	3	4	2	NONE	3	12	75.03	
	3	G395H/F290LP	S200A1	NRSRAPID	2	4	3	NONE	3	12	56.334	

Proposal 6987 - Observation 21 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Special Requirements	No Parallel Attachments
	Same Aperture PA 19, 20, 21 (V3 PAs differ)

Proposal 6987 - Observation 31 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 31: FS Mosaic center/butterfly - Repeat of obs 21										Tue May 26 15:00:26 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 31:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	-ETA-CAR	RA: 10 45 3.5378 (161.2647408d)				Proper Motion RA: -11 mas/yr					
			Dec: -59 41 4.05 (-59.68446d)				Proper Motion Dec: 4.1 mas/yr					
			Equinox: J2000				Epoch of Position: 2016.0					
		Comments: Updated to use Gaia DR3 position. Gaia DR3 does not provide any proper motions for this target, so we have adopted UCAC4 proper mostion but with the Gaia DR3 position at 2016.0 as the reference. The UCAC4 motion looks real, but it isn't clear if it was measuring the central star or some other ejected material. The uncertainty as to whether or not to include the proper motion probably adds ~ 0.1" to the final positional uncertainty. Category=Star Description=[Eta Carinae stars] Extended=YES										
Template	TA Method		HFF Readout Mode				Slit			Subarray		
	NONE		false				S200A1			SUBS200A1		
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	3	18	10.0		0.0		0.0		0.0		ROW_ORDER	
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Ex p	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F070LP	S200A1	NRSRAPID	4	4	1	NONE	3	12	93.726	
	2	G235H/F170LP	S200A1	NRSRAPID	3	4	2	NONE	3	12	75.03	
	3	G395H/F290LP	S200A1	NRSRAPID	2	4	3	NONE	3	12	56.334	

Proposal 6987 - Observation 31 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Special Requirements	Aperture PA Range 223.84190369 to 260.84190369 Degrees (V3 85.0 to 122.0) No Parallel Attachments
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Proposal 6987 - Observation 41 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Observation	Proposal 6987, Observation 41: FS Mosaic center/butterfly - Repeat of obs 31											Tue May 26 15:00:26 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 41:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	-ETA-CAR	RA: 10 45 3.5378 (161.2647408d) Dec: -59 41 4.05 (-59.68446d) Equinox: J2000				Proper Motion RA: -11 mas/yr Proper Motion Dec: 4.1 mas/yr Epoch of Position: 2016.0					
	Comments: Updated to use Gaia DR3 position. Gaia DR3 does not provide any proper motions for this target, so we have adopted UCAC4 proper mostion but with the Gaia DR3 position at 2016.0 as the reference. The UCAC4 motion looks real, but it isn't clear if it was measuring the central star or some other ejected material. The uncertainty as to whether or not to include the proper motion probably adds ~ 0.1" to the final positional uncertainty.											
	Category=Star											
	Description=[Eta Carinae stars] Extended=YES											
Template	TA Method		HFF Readout Mode				Slit			Subarray		
	NONE		false				S200A1			SUBS200A1		
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	3	18	10.0		0.0		0.0		0.0		ROW_ORDER	
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Ex p	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F070LP	S200A1	NRSRAPID	4	4	1	NONE	3	12	93.726	
	2	G235H/F170LP	S200A1	NRSRAPID	3	4	2	NONE	3	12	75.03	
	3	G395H/F290LP	S200A1	NRSRAPID	2	4	3	NONE	3	12	56.334	

Proposal 6987 - Observation 41 - Unveiling Eta Carinae's Eruptive Mass-Loss History with JWST Spectroscopy

Special Requirements	Aperture PA Range 223.84190369 to 260.84190369 Degrees (V3 85.0 to 122.0) No Parallel Attachments
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